

Лабораторная работа

Номер 3

Андрюшин Н. С.

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Российский университет дружбы народов, Москва, Россия

Информация

- Андрюшин Никита Сергеевич
- Студент
- Российский университет дружбы народов

Изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.


```
C:\Users\mega>ipconfig

Настройка протокола IP для Windows

Адаптер Ethernet Radmin VPN:

    DNS-суффикс подключения . . . . . : 
    IPv6-адрес . . . . . : fdfd::1a29:ea1
    Локальный IPv6-адрес канала . . . : fe80::f424:692c:f94b:d959%28
    IPv4-адрес . . . . . : 26.41.14.161
    Маска подсети . . . . . : 255.0.0.0
    Основной шлюз . . . . . : 26.0.0.1

Адаптер Ethernet Ethernet 3:

    DNS-суффикс подключения . . . . . : 
    Локальный IPv6-адрес канала . . . : fe80::5ce0:589d:3b4:4b8a%19
    IPv4-адрес . . . . . : 192.168.56.1
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз . . . . . : 

Адаптер беспроводной локальной сети Подключение по локальной сети* 1:

    Состояние среды . . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . : 

Адаптер беспроводной локальной сети Подключение по локальной сети* 2:

    Состояние среды . . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . : 

Адаптер Ethernet outline-tap0:

    Состояние среды . . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . : 

Адаптер беспроводной локальной сети Беспроводная сеть:

    DNS-суффикс подключения . . . . . : 
    Локальный IPv6-адрес канала . . . : fe80::35bc:d3f:6de:4b96%18
    IPv4-адрес . . . . . : 10.117.220.78
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз . . . . . : 10.117.220.134
```

Рис. 1: ipconfig

```
Адаптер беспроводной локальной сети Беспроводная сеть:
DNS-суффикс подключения . . . . . : 
Описание. . . . . : Killer(R) Wi-Fi 6E AX1675i 160MHz Wireless Network Adapter (211NGW)
Физический адрес. . . . . : 74-04-F1-D0-D0-45
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::35bc:d3f:6de:4b96%18(Основной)
IPv4-адрес. . . . . : 10.117.220.78(Основной)
Маска подсети . . . . . : 255.255.255.0
Аренда получена. . . . . : 11 октября 2025 г. 13:12:30
Срок аренды истекает. . . . . : 11 октября 2025 г. 16:31:34
Основной шлюз. . . . . : 10.117.220.134
DHCP-сервер. . . . . : 10.117.220.134
IAID DHCPv6 . . . . . : 175375601
DUID клиента DHCPv6 . . . . . : 00-01-00-01-2E-E6-8E-7E-74-04-F1-D0-D0-45
DNS-серверы. . . . . : 10.117.220.134
NetBios через TCP/IP. . . . . : Включен
```

Рис. 2: ipconfig /all

Выбор соединения

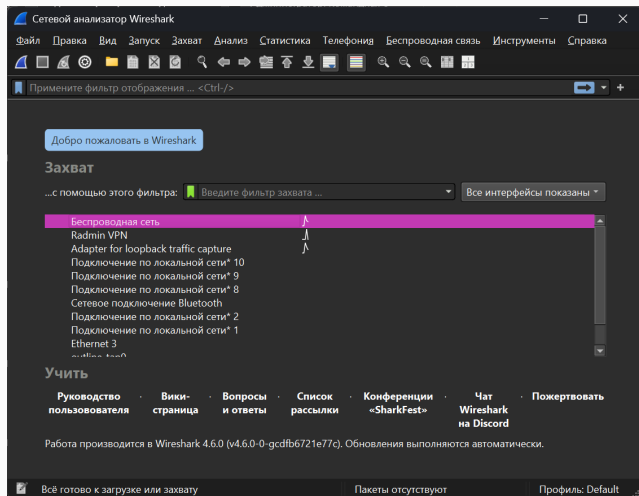


Рис. 3: Выбор соединения

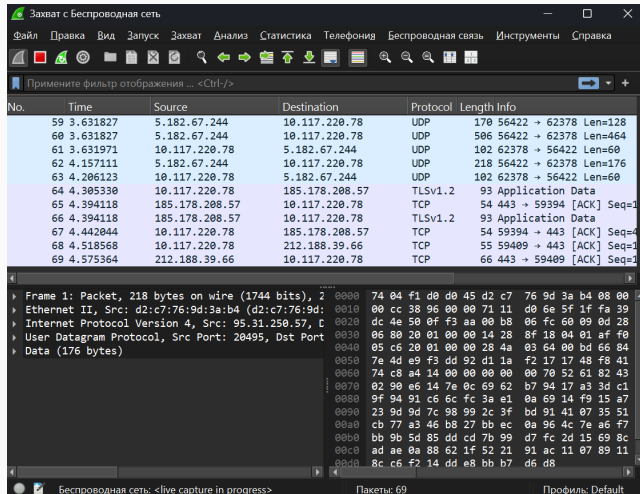


Рис. 4: Wireshark

```
C:\Users\mega_>ping 10.117.220.134

Обмен пакетами с 10.117.220.134 по 32 байтами данных:
Ответ от 10.117.220.134: число байт=32 время=21мс TTL=64
Ответ от 10.117.220.134: число байт=32 время=15мс TTL=64
Ответ от 10.117.220.134: число байт=32 время=6мс TTL=64
Ответ от 10.117.220.134: число байт=32 время=14мс TTL=64

Статистика Ping для 10.117.220.134:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 6мсек, Максимальное = 21 мсек, Среднее = 14 мсек
```

Рис. 5: ping

No.	Time	Source	Destination	Protocol	Length	Info
→	472 60.836202	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=86
←	473 60.857234	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=86
	475 61.849707	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=87
	476 61.864719	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=87
	494 62.862121	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=88
	495 62.868760	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=88
	502 63.876617	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=89
	503 63.890386	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=89

Рис. 6: icmp

Анализ исходящего пакета

```
Epoch Arrival Time: 1760187218.767445000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 866.353000 milliseconds]
[Time since reference or first frame: 1 minute, 836.202000 milliseconds]
Frame Number: 472
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
▼ Ethernet II, Src: Intel_d0:d0:45 (74:04:f1:d0:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
  ▼ Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
    .... 1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  ▼ Source: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
    .... 0 .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
```

Рис. 7: Анализ исходящего пакета

Анализ входящего пакета

```
▼ Frame 473: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{B7D3937E-0351-498A-B35E-CB1731CF0CFA}
  Section number: 1
  ▶ Interface id: 0 (\Device\NPF_{B7D3937E-0351-498A-B35E-CB1731CF0CFA})
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 11, 2025 15:53:38.788477000 RTZ 2 (зима)
  UTC Arrival Time: Oct 11, 2025 12:53:38.788477000 UTC
  Epoch Arrival Time: 1760187218.788477000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 21.032000 milliseconds]
  [Time delta from previous displayed frame: 21.032000 milliseconds]
  [Time since reference or first frame: 1 minute, 857.234000 milliseconds]
  Frame Number: 473
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:icmp:data]
  Character encoding: ASCII (0)
  [Coloring Rule Name: ICMP]
  [Coloring Rule String: icmp || icmpv6]
  ▼ Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
    ▼ Destination: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
      .....0. .... = LG bit: Globally unique address (factory default)
      .....0. .... = IG bit: Individual address (unicast)
    ▼ Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
      .....1. .... = LG bit: Locally administered address (this is NOT the factory default)
```

Рис. 8: Анализ входящего пакета

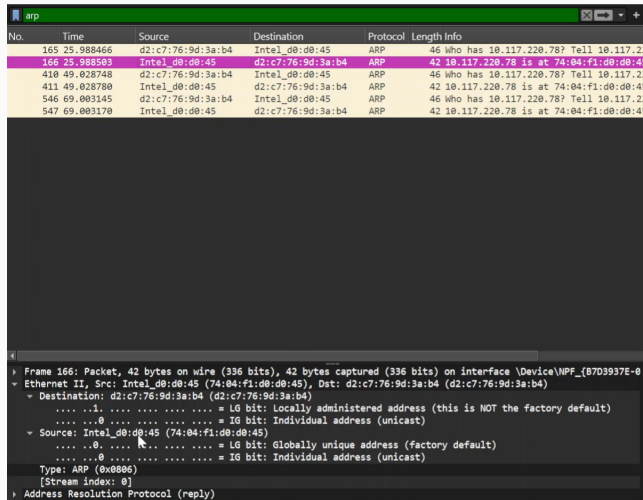
No.	Time	Source	Destination	Protocol	Length	Info
165	25.988466	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.22
166	25.988503	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:45
410	49.028748	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.22
411	49.028780	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:45
546	69.003145	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.22
547	69.003170	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:45


```

> Frame 165: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{B7D3937E-0
> Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
  > Destination: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  > Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
    ....1. .... = LG bit: Locally administered address (this is NOT the factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: 802.1Q Virtual LAN (0x8100)
  [Stream index: 0]
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
> Address Resolution Protocol (request)

```

Второй arp пакет



No.	Time	Source	Destination	Protocol	Length	Info
165	25.988466	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.2
166	25.988503	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:4
410	49.028748	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.2
411	49.028780	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:4
546	69.003145	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.2
547	69.003170	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:4

Frame 166: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{B7D3937E-0}

Ethernet II, Src: Intel_d0:d0:45 (74:04:f1:d0:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

- Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
 -1. = LG bit: Locally administered address (this is NOT the factory default)
 -0. = IG bit: Individual address (unicast)
- Source: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)

Type: ARP (0x0806)
[Stream index: 0]

Address Resolution Protocol (reply)

Рис. 10: Второй arp пакет

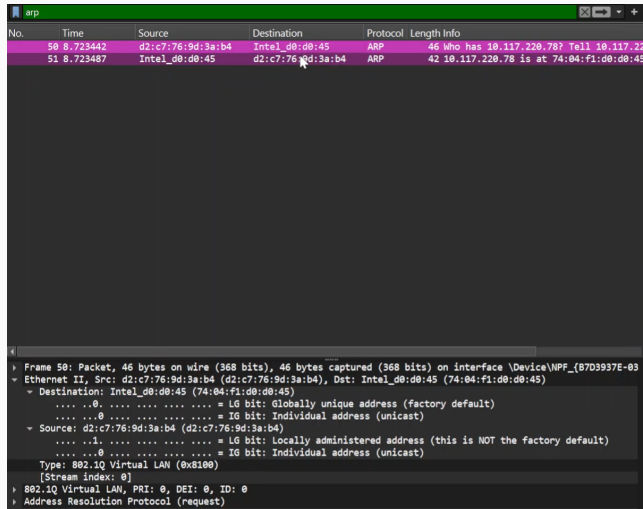
```
C:\Users\mega_>ping ya.ru

Обмен пакетами с ya.ru [77.88.44.242] с 32 байтами данных:
Ответ от 77.88.44.242: число байт=32 время=37мс TTL=54
Ответ от 77.88.44.242: число байт=32 время=38мс TTL=54
Ответ от 77.88.44.242: число байт=32 время=109мс TTL=54
Ответ от 77.88.44.242: число байт=32 время=62мс TTL=54

Статистика Ping для 77.88.44.242:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 37мсек, Максимальное = 109 мсек, Среднее = 61 мсек
```

Рис. 11: ping ya.ru

arp пакеты



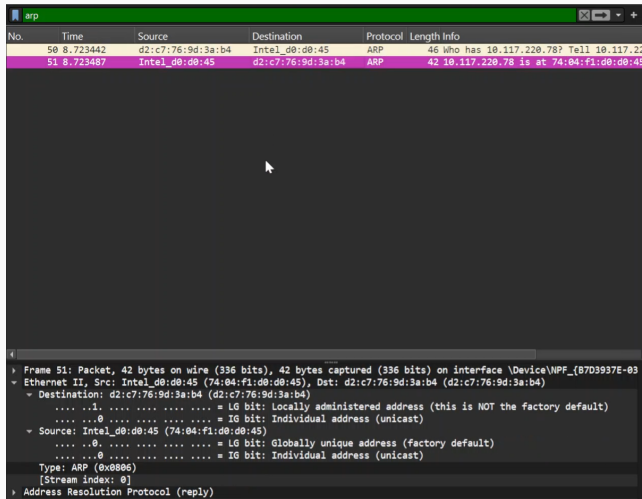
No.	Time	Source	Destination	Protocol	Length	Info
50	8.723442	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.220.78
51	8.723487	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:45

Frame 50: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{B7D3937E-03}

- Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
 - Destination: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
 - ...0... = LG bit: Globally unique address (factory default)
 - ...0... = IG bit: Individual address (unicast)
 - Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
 - ...1... = LG bit: Locally administered address (this is NOT the factory default)
 - ...0... = IG bit: Individual address (unicast)
 - Type: 802.1Q Virtual LAN (0x8100)
 - [Stream index: 0]
- 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
- Address Resolution Protocol (request)

Рис. 12: arp пакеты

Второй arp пакет



The image shows a Wireshark packet capture window titled 'arp'. The packet list pane shows two packets. Packet 51 is selected, and its details pane is expanded to show the Ethernet II and ARP sections.

No.	Time	Source	Destination	Protocol	Length	Info
50	8.723442	d2:c7:76:9d:3a:b4	Intel_d0:d0:45	ARP	46	Who has 10.117.220.78? Tell 10.117.220.78
51	8.723487	Intel_d0:d0:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:d0:d0:45

Frame 51: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{B7D3937E-03}

Ethernet II, Src: Intel_d0:d0:45 (74:04:f1:d0:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

- Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
 - ...1. = LG bit: Locally administered address (this is NOT the factory default)
 - ...0. = IG bit: Individual address (unicast)
- Source: Intel_d0:d0:45 (74:04:f1:d0:d0:45)
 - ...0. = LG bit: Globally unique address (factory default)
 - ...0. = IG bit: Individual address (unicast)

Type: ARP (0x0806)
[Stream index: 0]
Address Resolution Protocol (reply)

Рис. 13: Второй arp пакет

Исходящий пакет icmp

No.	Time	Source	Destination	Protocol	Length	Info
→ 24	3.422380	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=94
← 25	3.458802	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=94
← 28	4.438699	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=95
← 29	4.477199	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=95
← 31	5.453279	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=96
← 43	5.562230	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=96
← 45	6.467244	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=97
← 46	6.529173	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=97

Frame 24: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{B7D3937E-...}

Section number: 1

- Interface id: 0 (\Device\NPF_{B7D3937E-B351-498A-B35E-CB1731CF8CFA})
- Encapsulation type: Ethernet (1)
- Arrival Time: Oct 11, 2025 16:02:45.176440000 RTZ 2 (зима)
- UTC Arrival Time: Oct 11, 2025 13:02:45.176440000 UTC
- Epoch Arrival Time: 1768187765.176440000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 13.556000 milliseconds]
- [Time since reference or first frame: 3.422300000 seconds]
- Frame Number: 24
- Frame Length: 74 bytes (592 bits)
- Capture Length: 74 bytes (592 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:icmp:data]
- Character encoding: ASCII (0)
- [Coloring Rule Name: ICMP]
- [Coloring Rule String: icmp || icmpv6]

↳ Ethernet II, Src: Intel_d8:d0:45 (74:04:f1:d8:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

- ↳ Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
 -1. = LG bit: Locally administered address (this is NOT the factory default)
 -0. = IG bit: Individual address (unicast)
- ↳ Source: Intel_d8:d0:45 (74:04:f1:d8:d0:45)

Входящий пакет icmp

No.	Time	Source	Destination	Protocol	Length	Info
→ 4	0.133364	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=101/25856, ttl=128 (reply in 5)
← 5	0.168320	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=101/25856, ttl=54 (request in 4)
→ 9	1.146421	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=102/26112, ttl=128 (reply in 10)
→ 10	1.198248	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=102/26112, ttl=54 (request in 9)
→ 30	2.157817	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=103/26368, ttl=128 (reply in 31)
→ 31	2.186282	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=103/26368, ttl=54 (request in 30)
→ 34	3.169837	10.117.220.78	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=104/26624, ttl=128 (reply in 35)
→ 35	3.228482	77.88.44.242	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=104/26624, ttl=54 (request in 34)

▼ Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{B7D3937E-0351-498A-B35E-CB1731CF0CFA}, id 0

Section number: 1

▶ Interface id: 0 (\Device\NPF_{B7D3937E-0351-498A-B35E-CB1731CF0CFA})

Encapsulation type: Ethernet (1)

Arrival Time: Oct 11, 2025 17:10:15.263929000 RTZ 2 (smm)

UTC Arrival Time: Oct 11, 2025 14:10:15.263929000 UTC

Epoch Arrival Time: 1768191815.263929000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 34.956000 milliseconds]

[Time delta from previous displayed frame: 34.956000 milliseconds]

[Time since reference or first frame: 168.320000 milliseconds]

Frame Number: 5

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:icmp:data]

Character encoding: ASCII (0)

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

▼ Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_00:d0:d0:45 (74:04:f1:d0:d0:45)

▶ Destination: Intel_00:d0:d0:45 (74:04:f1:d0:d0:45)

▶ Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

Type: IPv4 (0x0800)

[Stream index: 0]

▶ Internet Protocol Version 4, Src: 77.88.44.242, Dst: 10.117.220.78

▶ Internet Control Message Protocol

Рис. 15: Входящий пакет icmp



Рис. 16: Подключение по http

Исходящий http пакет

No.	Time	Source	Destination	Protocol	Length	Info
88	3.469964	10.117.220.78	188.184.67.127	HTTP	355	GET / HTTP/1.1
94	4.660389	188.184.67.127	10.117.220.78	HTTP	932	HTTP/1.1 200 OK (text/html)
154	4.813872	10.117.220.78	188.184.67.127	HTTP	405	GET / HTTP/1.1
161	4.921661	188.184.67.127	10.117.220.78	HTTP	932	HTTP/1.1 200 OK (text/html)
186	5.402459	10.117.220.78	188.184.67.127	HTTP	420	GET /favicon.ico HTTP/1.1
190	5.485893	188.184.67.127	10.117.220.78	HTTP	308	HTTP/1.1 200 OK (image/vnd.microsoft)
680	62.216728	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
685	62.245875	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1183	73.277831	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
1187	73.314709	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1586	94.881268	10.117.220.78	212.188.39.73	HTTP	208	GET /connecttest.txt HTTP/1.1
1589	94.913184	212.188.39.73	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1935	109.803737	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
1943	109.853373	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
2407	120.450196	10.117.220.78	212.188.39.73	HTTP	208	GET /connecttest.txt HTTP/1.1
2413	120.492724	212.188.39.73	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
6835	727.799756	10.117.220.78	188.184.67.127	HTTP	465	GET /hypertext/MW/TheProject.html HT
6845	728.082427	188.184.67.127	10.117.220.78	HTTP	1104	HTTP/1.1 200 OK (text/html)
6874	731.185689	10.117.220.78	188.184.67.127	HTTP	495	GET /hypertext/DataSources/Top.html HT
6876	731.277065	188.184.67.127	10.117.220.78	HTTP	950	HTTP/1.1 200 OK (text/html)

Transmission Control Protocol, Src Port: 56203, Dst Port: 80, Seq: 1, Ack: 1, Len: 301
Source Port: 56203
Destination Port: 80
[Stream Index: 12]
[Stream Packet Number: 4]
Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 301]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2315743065
[Next Sequence Number: 302 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3158442879
0101 = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0xe842 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Timestamps
[Time since first frame in this TCP stream: 158.009000 milliseconds]
[Time since previous frame in this TCP stream: 355.000 microseconds]
SEQ/ACK analysis

Входящий http пакет

No.	Time	Source	Destination	Protocol	Length	Info
88	3.469964	10.117.220.78	188.184.67.127	HTTP	355	GET / HTTP/1.1
94	4.660389	188.184.67.127	10.117.220.78	HTTP	932	HTTP/1.1 200 OK (text/html)
154	4.813872	10.117.220.78	188.184.67.127	HTTP	405	GET / HTTP/1.1
161	4.921661	188.184.67.127	10.117.220.78	HTTP	932	HTTP/1.1 200 OK (text/html)
186	5.402459	10.117.220.78	188.184.67.127	HTTP	420	GET /favicon.ico HTTP/1.1
190	5.485893	188.184.67.127	10.117.220.78	HTTP	308	HTTP/1.1 200 OK (image/vnd.microsoft.i)
680	62.216728	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
685	62.245875	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1183	73.277831	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
1187	73.314709	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1586	94.881268	10.117.220.78	212.188.39.73	HTTP	208	GET /connecttest.txt HTTP/1.1
1589	94.913184	212.188.39.73	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
1935	109.803737	10.117.220.78	212.188.39.25	HTTP	208	GET /connecttest.txt HTTP/1.1
1943	109.853373	212.188.39.25	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
2407	120.450196	10.117.220.78	212.188.39.73	HTTP	208	GET /connecttest.txt HTTP/1.1
2413	120.492724	212.188.39.73	10.117.220.78	HTTP	241	HTTP/1.1 200 OK (text/plain)
6835	727.799756	10.117.220.78	188.184.67.127	HTTP	465	GET /hypertext/MwM/TheProject.html HTTP
6845	728.082427	188.184.67.127	10.117.220.78	HTTP	1104	HTTP/1.1 200 OK (text/html)
6874	731.185689	10.117.220.78	188.184.67.127	HTTP	495	GET /hypertext/DataSources/Top.html HT
6876	731.277065	188.184.67.127	10.117.220.78	HTTP	950	HTTP/1.1 200 OK (text/html)


```
Source Port: 80
Destination Port: 56203
[Stream Index: 12]
[Stream Packet Number: 6]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 876]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3158442879
[Next Sequence Number: 879 (relative sequence number)]
Acknowledgment Number: 392 (relative ack number)
Acknowledgment number (raw): 2315743366
0101 ... = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 249
[Calculated window size: 31872]
[Window size scaling factor: 128]
Checksum: 0xc12b [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
[client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (876 bytes)
```

Исходящий dns пакет

dns						
No.	Time	Source	Destination	Protocol	Length	Info
3	0.980996	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x84d6 A info.cern.ch
4	1.020694	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x84d6 A info.cern.ch
5	1.310404	10.117.220.134	10.117.220.78	DNS	112	Standard query response 0x84d6 A info.cern.ch
7	1.311479	10.117.220.78	10.117.220.134	DNS	77	Standard query 0xe97b A webafs902.cern.ch
8	1.319248	10.117.220.134	10.117.220.78	DNS	93	Standard query response 0xe97b A webafs902.cern.ch
9	1.319567	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA webafs902.cern.ch
10	1.362600	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA webafs902.cern.ch
14	1.483479	10.117.220.134	10.117.220.78	DNS	105	Standard query response 0x8704 AAAA webafs902.cern.ch

Frame 7: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Device\NPF_{B7D3937E-0B35-4000-8000-000000000000}

Ethernet II, Src: Intel_d0:d0:45 (74:04:f1:d0:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

Internet Protocol Version 4, Src: 10.117.220.78, Dst: 10.117.220.134

User Datagram Protocol, Src Port: 61668, Dst Port: 53

Source Port: 61668

Destination Port: 53

Length: 43

Checksum: 0xcdfb [unverified]

[Checksum Status: Unverified]

[Stream index: 1]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (35 bytes)

Domain Name System (query)

Рис. 19: Исходящий dns пакет

Входящий dns пакет

No.	Time	Source	Destination	Protocol	Length	Info
3	0.980996	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x84d6 A info.cern.ch
4	1.020694	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x84d6 A info.cern.ch
5	1.310404	10.117.220.134	10.117.220.78	DNS	112	Standard query response 0x84d6 A info.cern.ch
7	1.311479	10.117.220.78	10.117.220.134	DNS	77	Standard query 0xe97b A webafs902.cern.ch
8	1.319248	10.117.220.134	10.117.220.78	DNS	93	Standard query response 0xe97b A webafs902.cern.ch
9	1.319567	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA webafs902.cern.ch
10	1.362600	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA webafs902.cern.ch
14	1.483479	10.117.220.134	10.117.220.78	DNS	105	Standard query response 0x8704 AAAA webafs902.cern.ch

Frame 8: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface \Device\NPF_{B7D3937E-035...}

Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_d0:d0:45 (74:04:f1:d0:d0:45)

Internet Protocol Version 4, Src: 10.117.220.134, Dst: 10.117.220.78

User Datagram Protocol, Src Port: 53, Dst Port: 61668

Source Port: 53

Destination Port: 61668

Length: 93

Checksum: 0x55f1 [unverified]

[Checksum Status: Unverified]

[Stream Index: 1]

[Stream Packet Number: 2]

[Timestamps]

UDP payload (51 bytes)

Domain Name System (response)

Рис. 20: Входящий dns пакет

Исходящий quic пакет

No.	Time	Source	Destination	Protocol	Length	Info
6762	719.471867	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KP0), DCID=0d...
6763	719.495981	104.76.24.37	10.117.220.78	QUIC	66	Protected Payload (KP0)
6764	719.507133	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (KP0), DCID=0d...
6765	719.531381	104.76.24.37	10.117.220.78	QUIC	1288	Protected Payload (KP0)
6766	719.531381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KP0)
6767	719.531381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KP0)
6768	719.531381	104.76.24.37	10.117.220.78	QUIC	203	Protected Payload (KP0)
6769	719.531381	104.76.24.37	10.117.220.78	QUIC	105	Protected Payload (KP0)
6770	719.531381	104.76.24.37	10.117.220.78	QUIC	78	Protected Payload (KP0)
6771	719.531903	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KP0), DCID=0d...
6772	719.532140	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KP0), DCID=0d...
6773	719.532497	10.117.220.78	104.76.24.37	QUIC	77	Protected Payload (KP0), DCID=0d...
6774	719.533806	104.76.24.37	10.117.220.78	QUIC	1494	Protected Payload (KP0)
6775	719.535387	10.117.220.78	104.76.24.37	QUIC	75	Protected Payload (KP0), DCID=0d...
6776	719.537815	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KP0)
6777	719.537815	104.76.24.37	10.117.220.78	QUIC	1072	Protected Payload (KP0)
6778	719.537815	104.76.24.37	10.117.220.78	QUIC	77	Protected Payload (KP0)
6779	719.537815	104.76.24.37	10.117.220.78	QUIC	64	Protected Payload (KP0)
6780	719.539210	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KP0), DCID=0d...
6781	719.539383	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KP0), DCID=0d...
6782	719.539464	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KP0), DCID=0d...
6783	719.539527	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (KP0), DCID=0d...
6784	719.594855	104.76.24.37	10.117.220.78	QUIC	1494	Protected Payload (KP0)
6785	719.594855	104.76.24.37	10.117.220.78	QUIC	875	Protected Payload (KP0)
6786	719.595230	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KP0), DCID=0d...
Frame 6786: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_{B7D3937E-...}						
Ethernet II, Src: Intel_d0:d0:45 (74:04:f1:d0:d0:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)						
Internet Protocol Version 4, Src: 10.117.220.78, Dst: 104.76.24.37						
User Datagram Protocol, Src Port: 65115, Dst Port: 443						
Source Port: 65115						
Destination Port: 443						
Length: 39						
Checksum: 0x676d [unverified]						
[Checksum Status: Unverified]						
[Stream index: 102]						
[Stream Packet Number: 43]						
[Timestamps]						
UDP payload (31 bytes)						
QUIC IETF						

Входящий quic пакет

quic					
No.	Time	Source	Destination	Protocol	Length Info
6762	719.471867	10.117.220.78	104.76.24.37	QUIC	73 Protected Payload (KP0), DCID=0dt
6763	719.495981	104.76.24.37	10.117.220.78	QUIC	66 Protected Payload (KP0)
6764	719.507133	10.117.220.78	104.76.24.37	QUIC	74 Protected Payload (KP0), DCID=0dt
6765	719.531381	104.76.24.37	10.117.220.78	QUIC	1288 Protected Payload (KP0)
6766	719.531381	104.76.24.37	10.117.220.78	QUIC	1292 Protected Payload (KP0)
6767	719.531381	104.76.24.37	10.117.220.78	QUIC	1292 Protected Payload (KP0)
6768	719.531381	104.76.24.37	10.117.220.78	QUIC	203 Protected Payload (KP0)
6769	719.531381	104.76.24.37	10.117.220.78	QUIC	105 Protected Payload (KP0)
6770	719.531381	104.76.24.37	10.117.220.78	QUIC	78 Protected Payload (KP0)
6771	719.531903	10.117.220.78	104.76.24.37	QUIC	73 Protected Payload (KP0), DCID=0dt
6772	719.532140	10.117.220.78	104.76.24.37	QUIC	73 Protected Payload (KP0), DCID=0dt
6773	719.532497	10.117.220.78	104.76.24.37	QUIC	77 Protected Payload (KP0), DCID=0dt
6774	719.533806	104.76.24.37	10.117.220.78	QUIC	1494 Protected Payload (KP0)
6775	719.535387	10.117.220.78	104.76.24.37	QUIC	75 Protected Payload (KP0), DCID=0dt
6776	719.537815	104.76.24.37	10.117.220.78	QUIC	1292 Protected Payload (KP0)
6777	719.537815	104.76.24.37	10.117.220.78	QUIC	1072 Protected Payload (KP0)
6778	719.537815	104.76.24.37	10.117.220.78	QUIC	77 Protected Payload (KP0)
6779	719.537815	104.76.24.37	10.117.220.78	QUIC	64 Protected Payload (KP0)
6780	719.539210	10.117.220.78	104.76.24.37	QUIC	76 Protected Payload (KP0), DCID=0dt
6781	719.539383	10.117.220.78	104.76.24.37	QUIC	76 Protected Payload (KP0), DCID=0dt
6782	719.539464	10.117.220.78	104.76.24.37	QUIC	76 Protected Payload (KP0), DCID=0dt
6783	719.539527	10.117.220.78	104.76.24.37	QUIC	74 Protected Payload (KP0), DCID=0dt
6784	719.594855	104.76.24.37	10.117.220.78	QUIC	1494 Protected Payload (KP0)
6785	719.594855	104.76.24.37	10.117.220.78	QUIC	875 Protected Payload (KP0)
6786	719.595230	10.117.220.78	104.76.24.37	QUIC	73 Protected Payload (KP0), DCID=0dt

Frame 6784: Packet, 1494 bytes on wire (11952 bits), 1494 bytes captured (11952 bits) on interface \Device\NPF_{B...}

Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_d0:d0:45 (74:04:f1:d0:d0:45)

Internet Protocol Version 4, Src: 104.76.24.37, Dst: 10.117.220.78

User Datagram Protocol, Src Port: 443, Dst Port: 65115

Source Port: 443

Destination Port: 65115

Length: 1460

Checksum: 0xad2a [unverified]

[Checksum Status: Unverified]

[Stream index: 102]

[Stream Packet Number: 41]

[Timestamps]

UDP payload (1452 bytes)

QUIC IETF

Рис. 22: Входящий quic пакет

No.	Time	Source	Destination	Protocol	Length	Info
3	0.708788	10.117.220.78	188.184.67.127	TCP	66	56449 → 80 [SYN] Seq=0 Win=65535 Len=
4	0.961430	10.117.220.78	188.184.67.127	TCP	66	56451 → 80 [SYN] Seq=0 Win=65535 Len=
5	0.966938	188.184.67.127	10.117.220.78	TCP	60	80 → 56449 [SYN, ACK] Seq=0 Ack=1 Win=
6	0.967138	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=1 Ack=1 Win=6528
7	0.967892	10.117.220.78	188.184.67.127	HTTP	491	GET / HTTP/1.1
8	1.229785	188.184.67.127	10.117.220.78	TCP	66	80 → 56451 [SYN, ACK] Seq=0 Ack=1 Win=
9	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=1 Ack=438 Win=31
10	1.229785	188.184.67.127	10.117.220.78	HTTP	250	HTTP/1.1 304 Not Modified
11	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [FIN, ACK] Seq=197 Ack=438
12	1.230013	10.117.220.78	188.184.67.127	TCP	54	56451 → 80 [ACK] Seq=1 Ack=1 Win=6528
13	1.230144	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=438 Ack=198 Win=
14	1.230951	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [FIN, ACK] Seq=438 Ack=198
15	1.536721	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=198 Ack=439 Win=
16	1.741871	10.117.220.78	188.184.67.127	TLSv1.2	78	Application Data
17	1.741948	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [FIN, ACK] Seq=25 Ack=1 W
18	2.070672	188.184.67.127	10.117.220.78	TLSv1.2	78	Application Data
19	2.070672	188.184.67.127	10.117.220.78	TCP	54	443 → 56447 [FIN, ACK] Seq=25 Ack=26
20	2.070800	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [RST, ACK] Seq=26 Ack=25
21	2.143890	188.184.67.127	10.117.220.78	TCP	54	[TCP Retransmission] 443 → 56447 [FIN
22	3.283370	188.184.67.127	10.117.220.78	TCP	66	80 → 56446 [SYN, ACK] Seq=0 Ack=1 Win=
23	3.565224	10.117.220.78	208.115.231.158	TCP	55	52022 → 443 [ACK] Seq=1 Ack=1 Win=254
24	3.663852	208.115.231.158	10.117.220.78	TCP	54	443 → 52022 [ACK] Seq=0 Ack=2 Win=501
35	3.664098	10.117.220.78	208.115.231.158	TCP	54	[TCP Dup ACK 23#1] [TCP ACKed unseen
38	3.714200	208.115.231.158	10.117.220.78	TCP	66	[TCP Previous segment not captured] 4


```

Sequence Number: 0      (relative sequence number)
Sequence Number (raw): 3046997065
[Next Sequence Number: 1      (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
- Flags: 0x002 (SYN)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0... = Congestion Window Reduced: Not set
  ....0. .... = ECN-Echo: Not set
  ....0. .... = Urgent: Not set
  ....0... = Acknowledgment: Not set
  ....0... = Push: Not set
  ....0... = Reset: Not set
  ....0... = Syn: Set
  ....0... = Fin: Not set
[TCP Flags: .....S.]
Window: 65535
[Calculated window size: 65535]
Checksum: 0xe721 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options (if any): Maximum segment size: No Operation (NOP), Window scale: No Operation (NOP), No Operati

```

syn + ack

No.	Time	Source	Destination	Protocol	Length	Info
3	0.708788	10.117.220.78	188.184.67.127	TCP	66	56449 → 80 [SYN] Seq=0 Win=65535 Len=0
4	0.961430	10.117.220.78	188.184.67.127	TCP	66	56451 → 80 [SYN] Seq=0 Win=65535 Len=0
5	0.966938	188.184.67.127	10.117.220.78	TCP	64	80 → 56449 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
6	0.967138	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=1 Ack=1 Win=65280
7	0.967892	10.117.220.78	188.184.67.127	HTTP	491	GET / HTTP/1.1
8	1.229785	188.184.67.127	10.117.220.78	TCP	66	80 → 56451 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
9	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=1 Ack=438 Win=318
10	1.229785	188.184.67.127	10.117.220.78	HTTP	250	HTTP/1.1 304 Not Modified
11	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [FIN, ACK] Seq=197 Ack=438 Win=0
12	1.230013	10.117.220.78	188.184.67.127	TCP	54	56451 → 80 [ACK] Seq=1 Ack=1 Win=65280
13	1.230144	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=438 Ack=198 Win=65535 Len=0
14	1.230951	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [FIN, ACK] Seq=438 Ack=198 Win=0
15	1.536721	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=198 Ack=439 Win=318
16	1.741871	10.117.220.78	188.184.67.127	TLSv1.2	78	Application Data
17	1.741948	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [FIN, ACK] Seq=25 Ack=1 Win=0
18	2.070672	188.184.67.127	10.117.220.78	TLSv1.2	78	Application Data
19	2.070672	188.184.67.127	10.117.220.78	TCP	54	443 → 56447 [FIN, ACK] Seq=25 Ack=26 Win=0
20	2.070800	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [RST, ACK] Seq=26 Ack=25 Win=0 Len=0
21	2.143890	188.184.67.127	10.117.220.78	TCP	54	[TCP Retransmission] 443 → 56447 [FIN, ACK] Seq=25 Ack=26 Win=0
22	3.283370	188.184.67.127	10.117.220.78	TCP	66	80 → 56446 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
23	3.565224	10.117.220.78	208.115.231.158	TCP	55	52022 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=0
24	3.663852	208.115.231.158	10.117.220.78	TCP	54	443 → 52022 [ACK] Seq=0 Ack=2 Win=501 Len=0
35	3.664098	10.117.220.78	208.115.231.158	TCP	54	[TCP Dup ACK 23#1] [TCP ACKED unseen seq=52022]
38	3.714280	208.115.231.158	10.117.220.78	TCP	66	[TCP Previous segment not captured] 443 → 52022 [ACK] Seq=0 Ack=2 Win=501 Len=0

Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1484550668
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1476315037
1000 = Header Length: 32 bytes (0)
Flags: 0x012 [SYN, ACK]
000 = Reserved: Not set
...0 = Accurate ECN: Not set
...0... = Congestion Window Reduced: Not set
...0... = ECN-Echo: Not set
....0... = Urgent: Not set
....1... = Acknowledgment: Set
.....0.. = Push: Not set
.....0.. = Reset: Not set
.....1.. = Syn: Set
.....0.. = Fin: Not set
[TCP Flags:A..S.]
Window: 32120
[Calculated window size: 32120]
Checksum: 0x43d3 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operat

No.	Time	Source	Destination	Protocol	Length	Info
3	0.708788	10.117.220.78	188.184.67.127	TCP	66	56449 → 80 [SYN] Seq=0 Win=65535 Len=0
4	0.961430	10.117.220.78	188.184.67.127	TCP	66	56451 → 80 [SYN] Seq=0 Win=65535 Len=0
5	0.966938	188.184.67.127	10.117.220.78	TCP	66	80 → 56449 [SYN, ACK] Seq=0 Ack=1 Win=
6	0.967138	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=1 Ack=1 Win=65280
7	0.967892	10.117.220.78	188.184.67.127	HTTP	491	GET / HTTP/1.1
8	1.229785	188.184.67.127	10.117.220.78	TCP	66	80 → 56451 [SYN, ACK] Seq=0 Ack=1 Win=
9	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=1 Ack=438 Win=318
10	1.229785	188.184.67.127	10.117.220.78	HTTP	250	HTTP/1.1 304 Not Modified
11	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [FIN, ACK] Seq=197 Ack=438 W
12	1.230013	10.117.220.78	188.184.67.127	TCP	54	56451 → 80 [ACK] Seq=1 Ack=1 Win=65280
13	1.230144	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=438 Ack=198 Win=6
14	1.230951	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [FIN, ACK] Seq=438 Ack=198 W
15	1.536721	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=198 Ack=439 Win=3
16	1.741071	10.117.220.78	188.184.67.127	TLSv1.2	78	Application Data
17	1.741948	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [FIN, ACK] Seq=25 Ack=1 W
18	2.070672	188.184.67.127	10.117.220.78	TLSv1.2	78	Application Data
19	2.070672	188.184.67.127	10.117.220.78	TCP	54	443 → 56447 [FIN, ACK] Seq=25 Ack=26 W
20	2.070800	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [RST, ACK] Seq=26 Ack=25 W
21	2.143890	188.184.67.127	10.117.220.78	TCP	54	[TCP Retransmission] 443 → 56447 [FIN,
22	3.283370	188.184.67.127	10.117.220.78	TCP	66	80 → 56446 [SYN, ACK] Seq=0 Ack=1 Win=
23	3.565224	10.117.220.78	208.115.231.158	TCP	55	52022 → 443 [ACK] Seq=1 Ack=1 Win=254
24	3.663852	208.115.231.158	10.117.220.78	TCP	54	443 → 52022 [ACK] Seq=0 Ack=2 Win=501
35	3.664098	10.117.220.78	208.115.231.158	TCP	54	[TCP Dup ACK 23#1] [TCP ACKed unseen s
38	3.714200	208.115.231.158	10.117.220.78	TCP	66	[TCP Previous segment not captured] 44


```

Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1476315037
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1484550669
0101 .... = Header Length: 20 bytes (5)
  Flags: 0x010 (ACK)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    ....0... = Congestion Window Reduced: Not set
    ....0... = ECH-Echo: Not set
    ....0... = Urgent: Not set
    ....1... = Acknowledgment: Set
    ....0... = Push: Not set
    ....0... = Reset: Not set
    ....0... = Syn: Not set
    ....0... = Fin: Not set
    [TCP Flags: .....A....]
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0xe715 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0

```

Хендшейк на графике потока

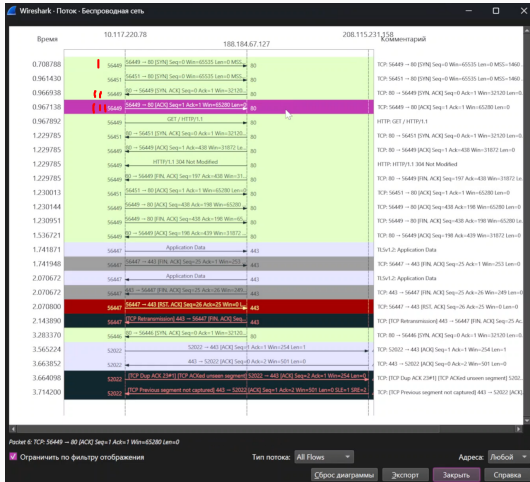


Рис. 26: Хендшейк на графике потока

В результате выполнения работы были получены навыки анализа пакетов в wireshark